Third Biweekly Project Progress Report

Project: PPT Outline Intelligent Generation and Content Completion System

Reporting Period (Sprint 3): 2026-04-26 – 2026-05-17

Prepared By: Ma Xiaolong (Team Leader, 2353814)

Team: Ma Xiaolong · Tang Chengyang · Zhang Ruiqi · Xie Jingcheng

1. Overall Progress Summary

Sprint 3 is the project's core implementation and end-to-end integration sprint. The team moved from "parallel module development" (Sprint 2 status) into module convergence, full-stack integration, and research-oriented evaluation, fully in line with the Updated Project Charter's WBS and Gantt Chart.

Headline outcomes this sprint:

- **WBS Section 2 (Core Module Development, 130 hrs budget)** advanced from In Progress to substantially complete (~90%); the backend asynchronous task state machine, RAG retrieval, prompt/Schema-constrained outline generation, and the frontend SSE streaming client all reached functional-prototype quality.
- **WBS Section 3 (Integration, Evaluation & Delivery, 50 hrs budget)** was started ahead of the original Gantt schedule and is now ~55% complete: a full user frontend, an admin frontend, and an end-to-end backend were integrated; the offline multi-model / RAG / DeepResearch evaluation pipeline was built and partially executed.
- **A production-blocking defect (PPT content-completion task hanging/deadlocking)** was diagnosed and fixed, materially improving end-to-end reliability.
- **The DeepResearch integration (WBS 3.2)** moved from Pending to In Improve, implemented on top of the Tavily advanced-search API and wired into the slide-batch generation pipeline alongside RAG.

Total hours invested this sprint: approximately 78 hours across the team (Sprint 1: 68 hrs, Sprint 2: 64 hrs; the increase reflects the integration-and-debugging workload of this phase).

The project remains on track for the firmly scheduled final delivery date of June 14, 2026.

2. Key Achievements This Sprint (RBS-aligned)

Structured Outline Generation (RBS-1):

- JSON-Schema-constrained four-level outline generation operational (Draft-07 schema enforced via chat_with_schema).
- Offline multi-model benchmark harness (Qwen/GLM/DeepSeek) built with Schema-compliance / ROUGE-L / BERTScore / latency / token-cost metrics.

Knowledge Retrieval & Content Completion (RBS-2):

- pgvector cosine-similarity RAG retrieval with outline-node-anchored queries, semantic chunking, and multi-format parsing complete.
- DeepResearch (Tavily advanced search) integrated and merged with RAG into the slide-generation

context.

- Content condensation pipeline (RAG + Web context → presentation-ready slide content with citation markers) operational.

End-to-End Evaluation System (RBS-3):

- Automated comparison script for LLM-only vs. LLM+RAG vs. LLM+RAG+DeepResearch built; preliminary results confirm RAG improves factual accuracy, DeepResearch adds domain-specific value.
- Likert-5 blind human-evaluation procedure designed and started.

Integration & Reliability (WBS 3.3):

- User frontend + admin frontend + backend + async workers integrated into a working end-to-end prototype.
- Production-blocking content-completion deadlock diagnosed and fixed (graceful degradation + thread-offloaded blocking calls + hardened async workflow).

3. Quantitative Analysis

3.1 Per-Member Hours by WBS Task — Sprint 3

WBS ID	Task	Owner	This-Sprint Hrs	Status	Completion %
2.1.2	Develop Session/Task asynchronous state machine logic	Ma Xiaolong	8	Completed	100%
2.1.3	Integrate OSS (MinIO) raw file storage interfaces	Ma Xiaolong	4	Completed	100%
3.3	System integration testing & blocking-bug fixing	Ma Xiaolong	8	In Progress	70%
3.4	Prototype deployment guide & technical documentation	Ma Xiaolong	3	In Progress	40%
2.2.2	Optimize outline generation prompts (JSON Schema)	Zhang Ruiqi	5	Completed	95%
2.2.3	Multi-model API integration & comparative testing	Zhang Ruiqi	11	In Progress	75%
2.3.2	pgvector vector indexing & similarity search	Tang Chengyang	4	Completed	100%
2.3.3	Multi-format document parsing plugins	Tang Chengyang	3	Completed	100%
3.1	End-to-end automated evaluation (ROUGE/BERTScore)	Tang Chengyang (+ Xie)	6	In Progress	70%
3.2	DeepResearch search workflow integration	Tang Chengyang	5	In Progress	65%
2.4.2	Frontend SSE streaming client	Xie Jingcheng	6	Completed	100%
2.4.3	Outline editing & preview interaction components	Xie Jingcheng	6	Completed	95%
3.x	Admin frontend + full-stack integration	Xie Jingcheng	4	In Progress	60%

WBS ID	Task	Owner	This-Sprint Hrs	Status	Completion %
	& stabilization				
—	Sprint review, planning & coordination	All (led by Ma Xiaolong)	~6	—	—

2.2 Group Total Hours per WBS Section (cumulative, all sprints)

WBS Section	Budget (hrs)	Cumulative Hrs Spent	Spent % of Budget	Task Completion %	Schedule Variance
1. Project Planning & Design	40	40	100%	100%	On schedule
2. Core Module Development	130	~118	91%	~90%	On schedule (slightly ahead on RAG/frontend)
3. Integration, Evaluation & Delivery	50	~24	48%	~55%	Ahead of plan (started earlier than the Gantt Integration phase 05-11)
Total	220	~182	83%	~80%	On / slightly ahead of schedule

4. Risk Analysis & Management

#	Risk	Category	Likelihood	Impact	Mitigation / Management Action	Status
R1	Multi-Model Comparison (WBS 2.2.3) ~1 week behind Gantt: full 15×3×3 benchmark runs not yet executed	Schedule	Medium	Medium	Harness already built; remaining work is execution-only. Parallelize API key provisioning; run benchmark in batch over the next sprint; buffer absorbed by the early start of Integration phase	Actively managed
R2	Async task hang / event-loop blocking causing content-completion deadlock	Technical	(Realized)	High	Resolved: blocking DashScope embedding offloaded to threads with single-call timeout; bounded retrieval-phase timeout with graceful degradation; per-task hard timeout + PEL recovery	Closed
R3	External data-source instability (Tavily / DashScope availability, rate limits, latency)	Environmental	Medium	Medium	Retry decorator on DeepSearch; bounded retrieval timeout that degrades to "continue without external context"; per-batch embedding timeout	Mitigated (by design)
R4	LLM JSON output not strictly Schema-compliant (malformed/partial JSON)	Technical	Medium	Medium	chat_with_schema injects Schema into system prompt + response_format=json_object; robust balanced-brace JSON extraction fallback in LLMClient._parse_json	Mitigated
R5	Single-developer concentration of integration knowledge (bus factor) on backend core	Interpersonal	Medium	Medium	STARTUP.md deployment guide authored; module boundaries documented; responsibilities mapped per Updated Charter RBS	Managed
R6	User trust / acceptance variance of AI-generated content across domains	Cultural fit	Low–Med	Medium	Citation markers + source-type labelling (report/RAG/Web) in exported content; human-evaluation track to surface domain gaps early	Monitoring
R7	Evaluation phase compression if benchmark + human-eval both slip	Schedule	Low	Medium	Evaluation harness reused across all three task tracks; human-eval scoring rubric finalized early; integration started ahead of plan to free buffer	Actively managed

Overall risk posture，The single realized high-impact risk (R2) has been closed. The only meaningful open schedule risk (R1, Multi-Model Comparison) is execution-bound, not design-bound, and is buffered by the early start of the Integration phase. Project risk is under control.

6. Upcoming Plan (Next Sprint)

Priority	Action	Owner	Target
P0	Execute full multi-model benchmark (15 × 3 models × 3 runs); deliver model-selection recommendation	Zhang Ruiqi	Close R1
P0	Complete DeepResearch query-refinement tuning + batch automated evaluation runs	Tang Chengyang	WBS 3.1/3.2
P1	End-to-end integration regression across full stack; finalize blocking-bug verification	Ma Xiaolong	WBS 3.3
P1	Complete automated end-to-end evaluation system + admin-side regression	Xie Jingcheng	WBS 3.1
P1	Finish human-evaluation collection; consolidate technical validation report	Tang Chengyang	RBS-3
P2	Package prototype deployment + technical documentation delivery	Ma Xiaolong + All	WBS 3.4

Ma Xiaolong

This-sprint hours: 23

Work completed (mapped to WBS 2.1, 3.3, 3.4):

- **Completed the Session/Task asynchronous state machine (WBS 2.1.2).** Finalized the Redis Stream consumer-group worker model: per-process multi-consumer naming for parallel XREADGROUP, an asyncio.Semaphore-bounded concurrent executor, a hard per-task timeout to prevent stuck workers, atomic PENDING → RUNNING DB transitions for idempotency, and PEL-recovery + orphan-message claiming so in-flight tasks survive process restarts. The session lifecycle (requirement_collection → outline_generation → outline_confirming → content_generation → content_confirming → completed) is enforced as a strictly forward-only state machine.
- **Integrated OSS / MinIO raw-file storage (WBS 2.1.3)** — moved from Pending (Sprint 2) to Completed; raw uploads are now persisted to object storage with the parsed-text pipeline downstream.
- **Diagnosed and fixed the production-blocking defect where PPT content completion would hang/deadlock.** Root cause was in the async task-processing path (event-loop blocking on synchronous embedding/IO calls + session handling). Fix: offloaded blocking DashScope embedding calls via asyncio.to_thread with a robust single-call timeout, added a bounded retrieval-phase timeout that degrades gracefully (continues without RAG/DeepSearch context rather than hanging), and hardened the slide-generation workflow. This materially improved end-to-end usability and reliability.
- **Led system integration testing (WBS 3.3)** across user frontend ↔ backend ↔ workers ↔ admin frontend, and led sprint review, planning and team coordination.
- **Drafted the prototype startup / deployment guide (STARTUP.md, WBS 3.4):** Dockerized PostgreSQL(pgvector)/Redis/MinIO dependencies and a reproducible dev bring-up.

Status / Next steps: State machine and OSS integration done. Continue end-to-end integration regression and begin packaging the technical-documentation deliverable.

Zhang Ruiqi

This-sprint hours: 16

Work completed (mapped to WBS 2.2):

- **Finalized JSON-Schema-constrained outline-generation prompt optimization (WBS 2.2.2):** the four-level topic → chapters → slides → points/notes outline now generates against a Draft-07 JSON Schema (outline_schema.json) with bounded array sizes and field-length constraints, materially improving structural soundness and hierarchical reasonableness.
- **Built an offline evaluation pipeline for the horizontal comparison of Qwen, GLM, and DeepSeek on the outline-generation task (WBS 2.2.3 / Multi-Model Comparison):**
  - 15 short-topic test cases with reference outlines.
  - Unified prompting via PromptLoader and Schema enforcement via chat_with_schema.
  - Metrics: Schema compliance rate, ROUGE-L, BERTScore, latency, token cost.
  - Extended LLMClient with chat_with_schema_and_usage to capture token usage for cost analysis.

Status / Next steps: Review and edge-case tuning of outline prompts; configure model API keys and execute the full outline_benchmark (15 samples × 3 models × 3 runs); deliver the comparison summary and the primary-model recommendation for production outline generation.

Tang Chengyang

This-sprint hours: 18

Work completed (mapped to WBS 2.3, 3.1, 3.2):

- Completed **pgvector vector indexing & similarity search (WBS 2.3.2):** cosine-distance (<=>) Top-K retrieval over document_chunks, score-threshold filtering, idempotent re-embedding, and outline-node-anchored query construction (chapter · slide title · points).
- **Completed multi-format document parsing plugins (WBS 2.3.3):** PDF (pdfplumber), DOCX (python-docx), MD/TXT, with text cleaning, SHA-256 dedup fingerprinting, and semantic paragraph-aware chunking with sliding-window overlap.
- **DeepResearch integration (WBS 3.2):** implemented multi-source web retrieval on the Tavily advanced search API, sharing the RetrievalResult structure with RAG and merged/deduplicated into the slide-batch generation context.
- **Multi-model horizontal comparison (jointly with Zhang Ruiqi):** evaluated Qwen / GLM / DeepSeek on Schema compliance, ROUGE-L, BERTScore, response time, token cost; produced model-selection recommendations.
- **RAG / DeepResearch enhancement comparison (WBS 3.1):** built an automated evaluation script (ROUGE-L, BERTScore, Schema compliance, custom PPT rules) comparing LLM-only vs. LLM+RAG vs. LLM+RAG+DeepResearch — confirmed RAG improves factual accuracy and DeepResearch adds value in domain-specific contexts.
- **Began human evaluation & final-report consolidation:** blind human evaluation on a Likert 5-point scale (structure, information density, factual accuracy, presentation readiness, overall satisfaction), consolidating all three task tracks into the technical validation report.

Status / Next steps: Finalize DeepResearch query-refinement tuning; complete batch automated-evaluation runs; finish human-evaluation collection and report integration.

Xie Jingcheng

This-sprint hours: 16

Work completed (mapped to WBS 2.4, 3.x):

- **Completed the frontend SSE streaming client (WBS 2.4.2):** EventSource-based subscription with token / progress / done / error event handling, per-session isolation to prevent cross-session interference, and automatic reconnection / task-resume on session switch — resolving the SSE display challenge flagged as a risk in Sprint 1.
- **Completed outline editing & preview interaction components (WBS 2.4.3):** the four-level outline editor and live preview.
- **Full-stack implementation & stabilization:** built and integrated the user frontend (login/registration, model/provider management, knowledge-base interaction, session/task workflow, PPT outline/content generation) and a separate admin frontend (dashboard, users, providers management); hardened backend support modules (LLM calling, prompt/schema loading, vector retrieval, task workers, export service, DB migration). Contributed to fixing the content-completion blocking issue and added startup documentation, improving end-to-end usability and reliability.
- **Began the end-to-end automated evaluation** harness (WBS 3.1) jointly with Tang Chengyang.

Status / Next steps: Outline-editor edge-case polish; complete the automated end-to-end evaluation system and admin-side integration regression.